

Computer Organization And Assembly Language

```
00000000: 01001101 01011010 10010000 00000000 00000011 00000000 MZ....
00000006: 00000000 00000000 00000100 00000000 00000000 00000000 .....
0000000c: 11111111 11111111 00000000 00000000 10111000 00000000 .....
00000012: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000018: 01000000 00000000 00000000 00000000 00000000 00000000 @.....
0000001e: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000024: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000002a: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000030: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000036: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000003c: 10000000 00000000 00000000 00000000 00001110 00011111 .....
00000042: 10111010 00001110 00000000 10110100 00001001 11001101 .....
00000048: 00100001 10111000 00000001 01001100 11001101 00100001 !..L.!
0000004e: 01010100 01101000 01101001 01110011 00100000 01110000 This p
00000054: 01110010 01101111 01100111 01110010 01100001 01101101 rogram
0000005a: 00100000 01100011 01100001 01101110 01101110 01101111 canno
00000060: 01110100 00100000 01100010 01100101 00100000 01110010 t be r
00000066: 01110101 01101110 00100000 01101001 01101110 00100000 un in
0000006c: 01000100 01001111 01010011 00100000 01101101 01101111 DOS mo
00000072: 01100100 01100101 00101110 00001101 00001101 00001010 de....
00000078: 00100100 00000000 00000000 00000000 00000000 00000000 $. ....
0000007e: 00000000 00000000 01010000 01000101 00000000 00000000 ..PE..
```

Lab Manual 03

Objectives:

1. Revision of different commands of debugger
2. Understanding of flags and their usage in conditional and iterative coding
3. Taking character input and Giving character and string output.

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Note: You are not required to handle the invalid inputs for task 1 to task 5.

Task 1:

Run the following codes on debugger and write down the status of flags:

i)

```
Mov ax,7FFFh Inc  
ax
```

ii)

```
Mov al,80h Dec  
al
```

iii)

```
Mov al, FFh  
Inc al
```

iv)

```
Mov ax,0000h Sub  
ax, 0001h
```

Task 2:

Take a single character input from user and display it.

Task 3:

Take a character input and tell whether it is capital or small.

→ Display “C” for capital letters and “s” for small letter.

Task 4:

Take two single digit numbers input and add them. Display correct output if the sum is less than 10, otherwise print “O”.

Task 5:

Fill an **8-byte array** in memory starting from offset 0508h (i.e., 0508h–050Fh in the DS segment) with random single-digit values for demonstration. Take a **singledigit number** (0–9) as input from the user and add it to each byte of the array. Finally, display the contents of memory before and after the addition so the effect of the operation can be clearly observed.

Task 6:

Describe functionality and working of following:

- DS Register
- CS Register
- IP Register
- SS Register
- Sign Flag
- Carry Flag
- Overflow Flag
- CMP instruction

Note: Task 6 have to be done on paper.

